

1

DESCRIPTION OF SPANISH VOWELS AND GUIDELINES FOR TEACHING THEM

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1.1 Introduction

This chapter describes the pronunciation of the five Spanish **vowels**, both stressed and unstressed, according to the context (see **phonetic context**) where they occur (e.g., whether they are in **open** or **closed syllables**). The chapter also deals with vowel groups, especially **diphthongs**, which are very frequent in Spanish. It highlights that in Spanish, phonological vowels are never realized as diphthongs (though in some cases they may be lengthened). The chapter addresses the main difficulties learners (regardless of their **L1**) have in mastering Spanish vowels, as well as teaching strategies that can be adopted to overcome them. We acknowledge that issues such as identity and other psycholinguistic aspects are also crucial to ways of speaking (including pronunciation), but have chosen an **articulatory phonetics** approach as the focus of this chapter. Also, the chapter is not tailored for use at a specific level of language since pronunciation is a long-term skill that should be taught in every stage of language learning (Bueno Hudson 2013).

1.2 Articulatory description of syllabic vowels

1.2.1 Definition

Vowels are speech **segments** characterized by two main features: they are **voiced** and constitute **syllable** nuclei (see **nucleus [in syllable structure]**). Syllables may consist of several segments, with their most essential part being a vowel, all syllables must contain a vowel, but not all syllables have **consonants**. Whereas in other languages segments such as **laterals** or **rhotics** can be syllable nuclei, in Spanish, only vowels can serve as a syllable nucleus. Diphthongs contain one non-syllabic vowel, or a **glide**, which is a non-nuclear vowel that is adjacent to the nuclear

vowel. Every vowel is a voiced sound and, as previously stated, can be pronounced in isolation (Gil 2007, 425). In this chapter, we will mainly focus on the description of standard Spanish vowels; however, colloquial and dialectal aspects of the pronunciation of vowels will also be addressed.

Phonologically speaking, the five vowel monophthongs /i, e, a, o, u/ **contrast** meaning in Spanish (e.g., *pipa*, *Pepa*, *papa*, *popa*, *pupa*), and therefore, constitute **phonemes** forming a triangle defined by the features [front, back] and [high, mid, low] (Figure 1.1). In terms of the horizontal axis, the front vowels are /i, e/, the back vowels are /u, o/, and /a/ is defined phonologically as neither front nor back, but rather central. Along the vertical axis, the vowels /i, u/ are high (or closed) vowels, /e, o/ are mid vowels, and /a/ is low (or open).

1.2.2 Phonetic features

In the articulation of the high and mid front vowels, [i] and [e], respectively, the **tongue** is raised and relatively **anterior** in the oral cavity, and closer to the **palate** in [i] than in [e]. For these vowels, the lips adopt a neutral position. The opening of the **pharynx** is more widely open for these front vowels than for [a]. In [u], the tongue gets close to the **velum** and the lips are more rounded, and in [o], the tongue gets close to the uvula and the top part of the pharynx, which is why these two vowels are classified as “back.” The lips are rounded in [u] and, to a lesser degree, in [o]. The pharynx area in [u] is also greater than in [a], which is also the case with [o], but in the lower pharyngeal area. The images in Figures 1.3 and 1.4 illustrate the features described in Figure 1.2 in detail by showing articulatory organs and positions.

In the articulation of [i] and [e], the tongue rises toward the palate and the front of the oral cavity while the lips remain unrounded (Figure 1.4). The space between the root of the tongue and the pharynx wall is wider the higher the tongue is. In

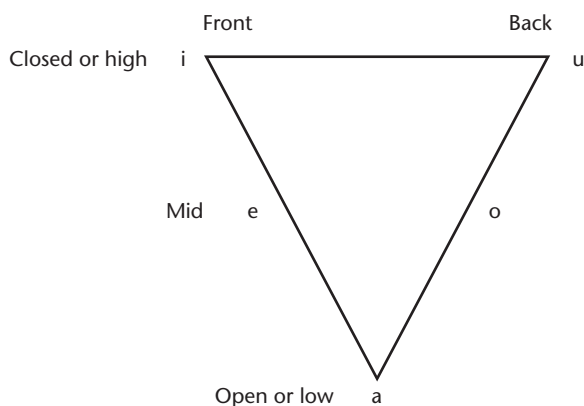


FIGURE 1.1 Schematic representation of the Spanish vowel system.

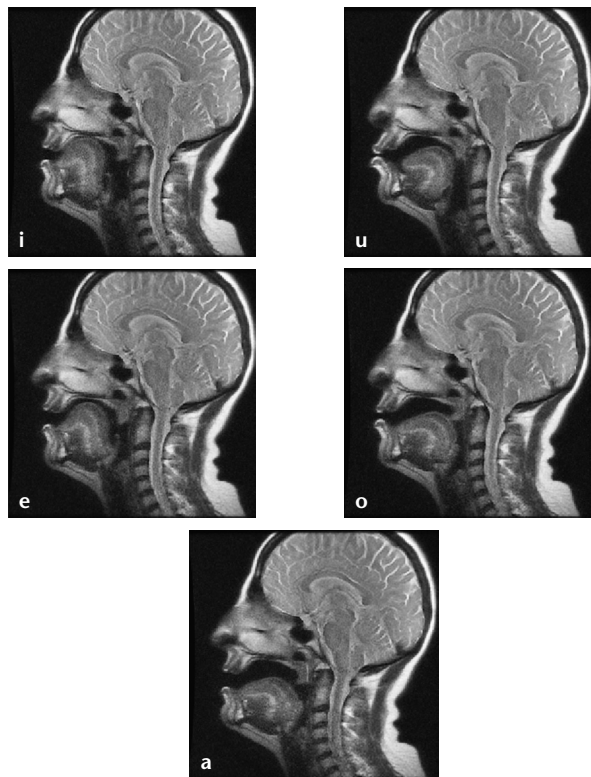


FIGURE 1.2 MRIs of Spanish vowels.

Source: Iribar (2012; 2013, 78). Reprinted by permission of *Estudios de Fonética Experimental*. Barcelona, Spain.

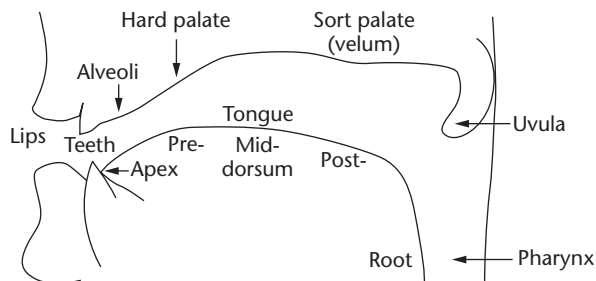


FIGURE 1.3 The main articulators involved in the production of vowels.

the articulation of [a], a low central vowel, the tongue remains low and neither front nor back, and the pharynx narrows considerably. In the articulation of [u] and [o], the tongue retracts to a **velar** or **uvular** position, respectively. The lips are more rounded and closer for [u] than for [o]. As already described for front vowels, the pharyngeal space is wider in high vowels.

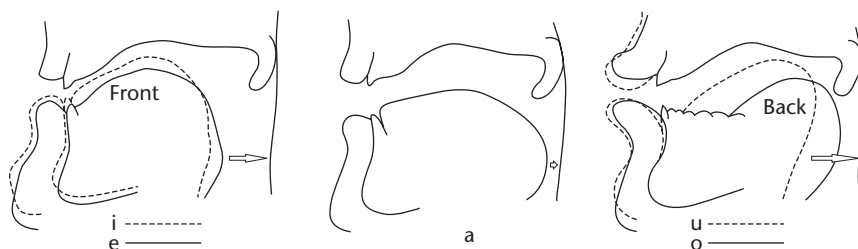


FIGURE 1.4 Articulatory diagrams of Spanish vowels showing tongue position, lip rounding, and pharyngeal width.

1.2.3 Phonetic variation

Spanish vowels are tense and short, and present little variation resulting from contextual phonetic processes, any variation that does exist is mostly due to adjacency to certain consonants or another vowel, the latter of which results in a diphthong. In Standard Spanish in particular, vowels do not undergo any phonological processes; for example, they are not reduced in unstressed position or dropped at the end of a word. The main types of contextual variation in the realization of the Spanish monophthong vowels are outlined in Section 1.2.3.

1.2.3.1 Height

Navarro Tomás (1918) described a series of degrees of vowel height for Castilian Spanish that depend on the context in which each vowel is pronounced, but in fact, little contextual variation exists. Thus, there are not truly closed and open variants of /e/, but rather it can be articulated slightly lowered or more opened. The same can be said regarding the other vowels, except for /a/, which varies slightly across the front–back dimension (Figure 1.5).

In terms of the small set of contextual variants, according to Navarro Tomás (1918), all vowels become more open in three contexts: 1. As part of a closed syllable, as in *til.de*, *ter.co*, *cor.to*, and *pul.so*, 2. When in contact with a trill [r], as in *irritar*, *remedio*, *perro*, *corre*, *ruta*, and *currar*, and 3. When preceding the sound [x], as in *hijo*, *queja*, *cojo*, and *lujo*. It should be noted that /e/ has some restrictions concerning the first rule, when it is followed by /m, n, s, d, θ/, as in *em.pe.zar*, *men.te*, *pes.ca.do*, *sed*, and *diez*, it does not become more open. In addition, /e/ usually becomes more open in the diphthong [ej̞] (e.g., *seis*) such that it can even be pronounced ['saj̞s] in informal speech, although the more frequent realization is ['sej̞s]. It should be noted that the diacritic under [e] indicates a slight lowering of the jaw, or a more open articulation (the arrows in Figure 1.5 indicate the tendency of non-low vowels to lower in the above-mentioned contexts), but one that is not as extreme as that of [ɛ]. The vowel /o/ also becomes more open in the diphthong [oi̞] (e.g., *voy*, *soy*); however, unlike /e/, /o/ has no exceptions in closed syllables.

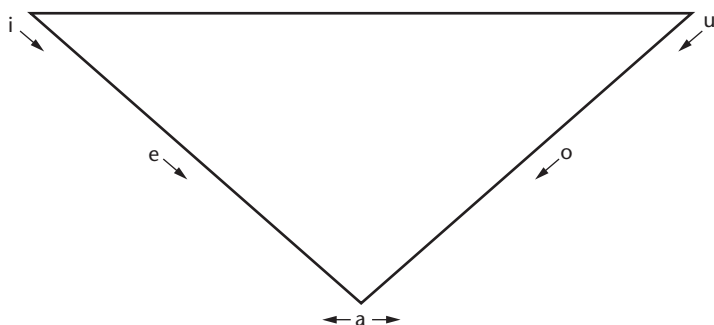


FIGURE 1.5 Context-dependent vocalic movements.

Furthermore, the low vowel /a/ can be realized as slightly fronted in contact with **palatal** consonants (e.g., *yaya*, *llamar*, *caña*), and usually becomes slightly more back before the consonant /x/ (e.g., *ajo*, *paja*), before [o, u, ʊ] (e.g., *ahora*, *baúl*, *causa*), and in a syllable closed by /l/ (e.g., *alma*, *calvo*).

Acoustically, no observable differences in vowel height exist due to context (Martínez Celdrán and Fernández Planas 2007, 183–188), but articulatory differences do exist and are observable through electropalatography. Figure 1.6 shows a diagram of an electropalatogram. It has eight rows indicating different areas of the palate where the tongue can make contact (i.e., different places of articulation): the end of the teeth, the alveoli, the hard palate, and the beginning of the velum. The degree of contact between the tongue and each region is shown through different colors that appear in the squares, complete contact is signaled by black squares, while different shades of gray indicate partial contact (dark gray = 50 percent, gray = 25 percent) and white indicates no contact.

Non-open vowels display more contacts than slightly open vowels, as seen in Figure 1.7. Comparing *tesis* and *seis* shows that the [e] of *tesis* has 14 squares with 100 percent, four with 50 percent and four with 25 percent in the palatal area. The [ɛ] of *seis* has eight squares with 100 percent, six with 50 percent and six with 25 percent. This difference is not large but there clearly is less contact in *seis*, indicating a certain degree of opening. Comparing *lista* and *tisis* demonstrates that the [i] in

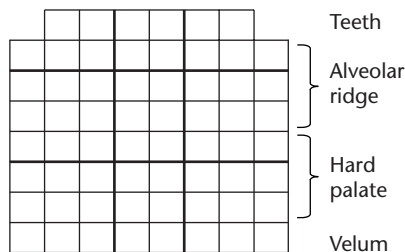


FIGURE 1.6 Diagram of an electropalatogram.

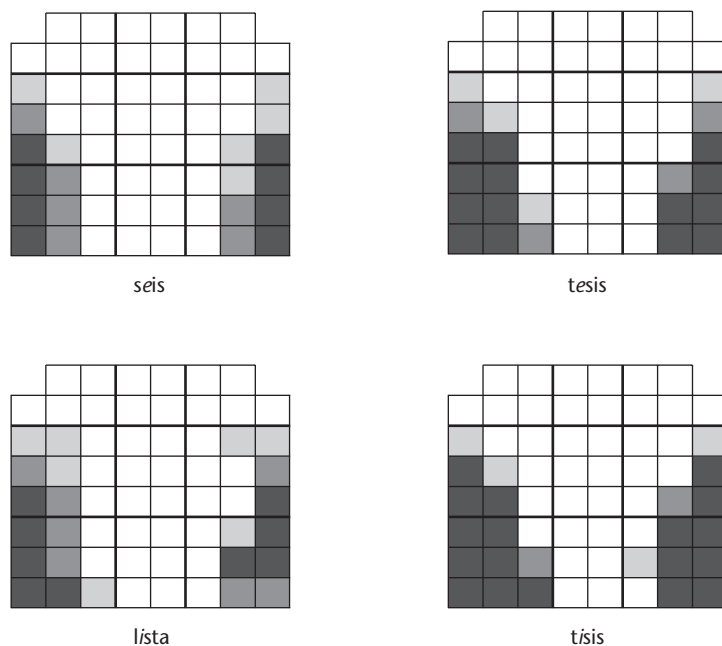


FIGURE 1.7 Electropalatograms showing the degree of vowel opening in productions of /e/ and /i/.

Source: adapted from Martínez Celdrán and Fernández Planas (2007, 187).

lista (in a closed syllable) shows less contact than the first [i] of *tísis* (in an open syllable). In sum, the images in Figure 1.7 illustrate that the more open the vowel, the less contact we see in electropalatograms.

1.2.3.2 Nasalization

There are no nasal vowel phonemes in Spanish, but vowels may become nasalized in syllables ending in a nasal consonant, and they are obligatorily nasalized when they are between nasals (see **nasal-stop class**); for example, in the utterance *en seguida vuelvo*, almost certainly the first vowel nasalizes, as is the case in *desempeñar* and *detente* as well. In *mañana no iré*, the nasalization (see **vowel nasalization**) of the three /a/ vowels is obligatory because they all fall between two nasal consonants (for further details on this topic, see Chapter 6 of this volume).

1.2.3.3 Dialectal variation

One of the most salient characteristics of vowel variation in Spanish is the existence of clearly open vowels in some dialects. In Caribbean, Eastern Andalusian and Murcian Spanish (among many other dialects), **coda** /s/ is either elided or aspirated

(for more on this topic, see Chapters 4 and 9). This elision (see **/s/-deletion/** aspiration (see **/s/-aspiration**) triggers the opening of the preceding vowel (e.g., *este* ['ɛh_ɪtɛ]). Moreover, in the plurals of these dialects, the elision/aspiration of /s/ and accompanying vowel opening trigger a long-distance **assimilation** process whereby all vowels in the word become open, which is a phenomenon known as *vowel harmony* (e.g., *otros hombres* ['ɔ_ɪtɾɔ.h'ɔm.brɛh]).

Vowel devoicing and elision in unstressed syllables (i.e., *vocales caedizas*) are also possible in various dialects of Spanish. This has been described in the Spanish spoken in the Mexican Plateau, Costa Rican Central Valley, Colombia, the Ecuadorian highlands, Peru, Bolivia, and Santiago del Estero in Argentina. This phenomenon is particularly common when vowels are in contact with **dental** sounds and/or are the final vowel(s) of a word (Lope Blanch 1983, 57–77), for example, *pues* [ps] and *dientes* ['diɛnts].

In addition to the nasalization processes that are common in all Spanish dialects, Caribbean Spanish and Andalusian Spanish vowels can nasalize in other contexts. In particular, they can nasalize when followed by a nasal and, in such cases, the nasal can even be elided (e.g., *tonto* ['tũ̃.ɲo]), especially when it is in final position (e.g., *pan* ['pãn] or ['pã]). At times, all the vowels in a word are nasalized due to vowel harmony, as in *empezar* [ẽm.pẽ.'sãr] (Vaquero Ramírez 1996).

1.2.4 Prosodic properties

Prosody deals with **suprasegmental features**, or those involving combinations of multiple sounds. In this subsection, we will specifically address pitch (i.e., fundamental frequency [F0]), vowel duration, and **intensity**. F0 is measured in Hertz (Hz), duration is measured in milliseconds (ms), and intensity is measured in decibels (dB).

F0, or what we perceive as *pitch*, is the main acoustic (see **acoustic phonetics**) correlate of prosodic events in Spanish. The F0 contour of a sentence is usually divided into the **nucleus** and **prenucleus**. The nucleus (or *tonema*) of an utterance includes the last stressed syllable and any post-tonic syllables, while all items coming before it form the pre-nucleus; for example, in the sentence *Inés vendrá mañana en coche*, *coche* constitutes the nucleus and the rest of the sentence is the pre-nucleus. Intonation studies have shown that most Spanish dialects displace F0 peaks to post-tonic syllables in the pre-nucleus (Sosa 1999, 193). This means that F0 begins to rise in the stressed syllable but usually peaks in the next syllable. In the nucleus, however, peaks are reached within the stressed syllable. Finally, in non-emphatic statements, peak height tends to decline across the utterance.

The most significant contextual variation tied to vowel duration in Spanish is found in words at the end of an utterance before a pause, or the nucleus, where the last syllable is longer than non-final syllables. Data from Fernández Planas and Martínez Celdrán (2003, 187) show that, in statements, nuclear vowel duration varies from 78 ms (non-final syllable) to 104 ms (final syllable), which is a 26 ms difference on average. In nuclear position of questions, vowel duration varies from 48 ms (non-final) to 83 ms (final), meaning a 35 ms difference on average. On the

other hand, in the prenucleus, contextual differences are below the duration **threshold**, which is estimated to be around 20 ms. In fact, implementing more involved metrics with the nuclear duration values presented shows that the threshold is only overcome in questions and not statements. As such, overall, we can conclude that, perceptually, duration does not play a significant role in vocalic differences. While this is the case, we can generalize that stressed syllables are slightly longer than unstressed syllables and that stressed syllables can be lengthened for different pragmatic functions (e.g., emphasis). Both of these phenomena can affect the length of vowels belonging to stressed syllables.

Intensity does not play a significant role in Spanish prosody, as differences in intensity between stressed and unstressed vowels are negligible and, unlike duration, intensity diminishes at the end of the utterance.

In Figure 1.8, the duration bar graph shows that the duration of the stressed vowel (3) of the first stressed word in the prenucleus (*guitarra*) is slightly higher than those of the pre-tonic and post-tonic syllables (2 and 4, respectively), but in the second stressed word (*toca*), the stressed vowel (6) is shorter than the post-tonic vowel (7). Duration increases in the nucleus (*paciencia*), where it begins its increase in the stressed vowel (10) and ultimately reaches its maximum value in the post-tonic (11) vowel, which is the last vowel of the utterance. Furthermore, a glance at the intensity bar graph tells us that it does not vary much and only declines slightly

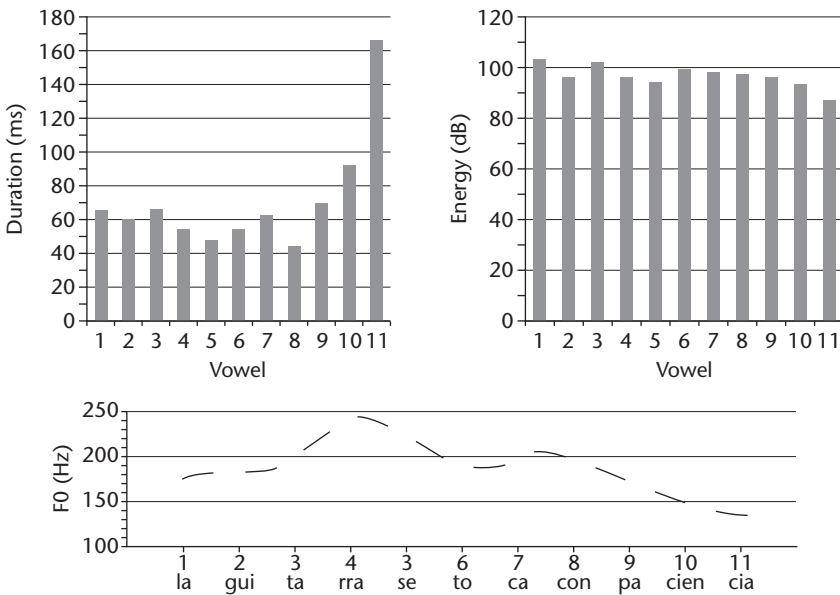


FIGURE 1.8 An example of duration (upper left panel), intensity (upper right panel), and F0 (bottom panel) of the vowels of the sentence *La guitarra se toca con paciencia*.

Source: adapted from AMPER (http://stel.ub.edu/labfon/amper/cast/ampercat_resultados_old.html).

at the end. Finally, the F0 contour demonstrates that, since the utterance is a statement, the prenucleus contains rises through stressed syllables and high targets in the post-tonic syllables, while in the nucleus, the contour falls to the end of the sentence, and therefore, the stressed syllable's level is higher than that of the post-tonic syllable (for an in depth treatment of prosody, see Chapter 8).

1.3 Non-syllabic vowels

Non-syllabic vowels are those that are not located in central position of the syllabic nucleus. They are called glides or semivowels and are considered different from syllabic, or nuclear (i.e., “true”), vowels, they are produced as transitions before or after nuclear vowels, but cannot be lengthened. On the other hand, speakers can increase or decrease the duration of regular vowels based on factors such as the level of formality of their speech. If a glide were to lengthen, it would turn into a true vowel. An example that illustrates the distinction between a nuclear vowel and a glide is the /i/ in *María* versus *Mariana*. The first name divides the sequence <ria> across two syllables (i.e., -rí.a), whereas in the second name, this sequence comprises one syllable (i.e., -ria-); phonetically, these translate to ['rí.a] and ['rí.a], respectively. In the first name, [i] is realized as a nuclear vowel in the syllable ['rí], with the following [a] being the nucleus of its own syllable, whereas in the second name, /i/ is produced as a glide [j] occurring before the syllabic nucleus [a] in the single syllable ['rí.a] (see Martínez Celdrán and Fernández Planas 2007, 161–170).

The combination of one or two non-syllabic vowels and a syllabic vowel in the same syllable is called a diphthong or **triphthong**, respectively (see MacLeod 2015). Spanish has 16 diphthongs. Some of them occur very often in speech for historical reasons. In the development of Spanish, Latin short mid vowels diphthongized in stressed position (e.g., $\text{ě} > \text{ie}$ [ie] and $\text{ō} > \text{ue}$ [ue]), which resulted in Spanish words such as *tierra*, *bien*, *pienso*, *puerto*, *bueno*, and *puedo*. In these cases, there are always two well-defined phonemes that can commute /i-/:/u-/ (e.g., *bien/buen*) or /-e-/:/-a/ (e.g., *cuento/cuanto*), that is, Spanish never diphthongizes a vowel phoneme, no matter how long it is. Often, when someone says something strange, the addressee asks ¿Quéééééé?! (where the repetition of <e> represents a long vowel); however, regardless of its length, the vowel of this word is never diphthongized. On the other hand, in English, final vowels often diphthongize; for example, the Spanish word *nacho* is adapted in American English as ['nɑ:.tʃoʊ]. In Spanish, diphthongs can be formed when the glides [j] or [w] precede the center of the nucleus (i.e., **rising diphthong**), as in the examples above like *bien* ['ie] or *bueno* ['ue], or when these glides occur after the center of the nucleus (i.e., **falling diphthong**), as in *causa* ['au], *leucocito* [eu], *estadounidense* [ou], *caiman* [ai], and *coito* ['oi]. A sequence of three vowels in which the center of the nucleus is flanked by glides is a triphthong, for example, *averiguáis* ['wai], *cambiéis* ['iei], *buey* ['uei], *miau* ['iau], and *guan* ['uau]. In diphthongs and triphthongs, the most open vowel is always the center of the syllable nucleus. When a sequence of two vowels are high, the second is usually

the center of the nucleus and the first one is a glide, as in *fui* ['ui], *cuidado* [ui], *ciudad* [iu], *viuda* ['iu], and *buitre* ['ui] (for further details on diphthongs, see Chapter 7).

A sequence of two vowels that belong to different syllables constitutes a **hiatus**, which is longer than a diphthong given that the two vowels maintain their typical durations, that is, a hiatus does not contain a glide. The most common type of hiatus consists of a high stressed vowel in contact with another vowel, as in the ['i.a] sequence of *había* and *María*; however, sometimes a hiatus undergoes compression and becomes a diphthong when forming compounds (e.g., ['ja] in *María-José*). Usually, hiatuses are formed by two adjacent vowels that are not closed (i.e., non-high vowels), as in *teatro* [e.'a], *maestro* [a.'e], *reo* ['e.o], and *roer* [o.'e]. Producing a hiatus as a diphthong may depend on factors such as language register (see **formal/informal register**), dialectal variety or speech rate; for example, the word *maestro*, which is stressed on the /e/, can have its stress shifted to the preceding /a/ and be pronounced in two syllables (i.e., ['maɛs.tɾo]). In informal speech, vowels can also become higher and produce a diphthong, as in *teatro* > *tíatro* ['ia] and *Joaquín* > *Juaquín* [uɪ]. In fact, it is quite common for hiatuses to be produced as diphthongs. Such compression is also common in poetry, where it receives the name **synalepha**; for example, in *la hermosa*, <la her> can form a single syllable. Garcilaso, a poet from the Renaissance, did not produce synalepha here because /h/ was still pronounced in his times, but all poets after his time would do so since the phoneme /h/ was absent in their speech.

When two adjacent vowels belong to different words, they diphthongize when one of the vowels is higher than the other. This is the case in *lo humano* [ou], *este invierno* [ɛi], *la inyección* [ai], and *la humanidad* [au].

Finally, when two adjacent vowels have the same quality, either one of the two vowels is elided (e.g., *lo horrible* [lɔ.'ri.βle]) or they merge, resulting in vowel lengthening (e.g., *lo horrible* [lɔ:.'ri.βle], *creer* ['krɐ:r]).

1.4 Comparisons with other languages

Cardinal vowels serve as a universal reference system representing all human articulatory possibilities in the production of any vowel. When the vowels of a language are depicted in a cardinal vowel chart, users can compare the vowels of that language to the general possibilities of the pronunciation of vowels in any world language. Figure 1.9 shows a vowel **formant chart** plotting the International Phonetic Alphabet's (IPA's) cardinal vowels, and places their theoretical values (Quilis 1980, 19) together with average values of Spanish vowels. The cardinal vowels represent extreme reference points in the articulation of vowels and Spanish vowels form a smaller, triangular space within the extremes. Additionally, the majority of Spanish vowels are much more open (i.e., lower) than cardinal ones; for example, the closed /o/ of many languages (e.g., German) may sound like /u/ to Spanish speakers.

Formant plots can also be used to compare L1 vowels to Spanish vowels. Average formant values coming from male speech are presented in Table 1.1, and Figure 1.10 illustrates the comparison of Spanish vowels to French and English vowels. In the plots, formant values are reported in the Bark unit (rather than Hz)

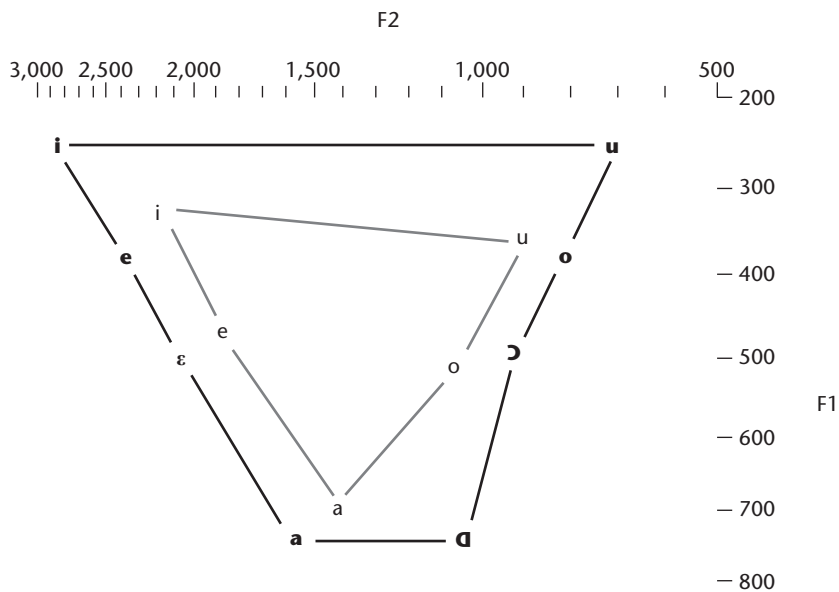


FIGURE 1.9 Spanish vowels versus cardinal vowels.

TABLE 1.1 Vowel formant values from a male Spanish speaker (Martínez Celdrán and Fernández Planas 2007, 175). Female values can be consulted in Martínez Celdrán and Fernández Planas (2007, 177)

Values in Hz	<i>i</i>	<i>e</i>	<i>a</i>	<i>o</i>	<i>u</i>
1st formant	313	457	699	498	349
2nd formant	2,200	1,926	1,471	1,070	877

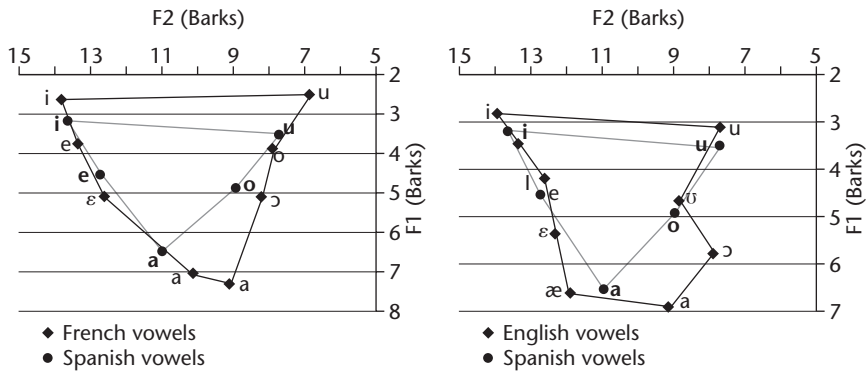


FIGURE 1.10 Normalized formant charts comparing Spanish vowels with those of French and American English.

because this scale gives normalized values (Hayward 2000, 142). Spanish vowels (Martínez Celdrán and Fernández Planas 2007, 175) are slightly more open than French vowels, except for /a/, which is higher than French /a/ (Landeracy and Renard 1977, 86; Quilis 1980, 18). This reflects the Spanish versus American-English comparison as well (Ladefoged 1993, 193). Examples of this are that English /ɪ/ is close to Spanish /e/ and English/ʊ/ is next to Spanish /o/.

1.4.1 *Previous studies on the second language acquisition of Spanish vowels by native English speakers*

Given that many readers of this book will be teaching Spanish vowels to native English speakers and that almost all previous work on **L2** Spanish vowels has been done using L1 English learners, in this section, we will summarize what previous research cites as some of the main challenges L1 English speakers face when learning Spanish vowels.

The vast majority of the literature focuses on difficulties native English learners of Spanish have in avoiding **vowel reduction** processes (e.g., Aldrich 2014; Cobb and Simonet 2015; Bland 2016), which is also a problem for native speakers of other languages (e.g., Dutch, as seen in Iruela 1997; for examples of other languages, see Section 1.5.3). Vowel reduction involves shortening and/or centralizing vowels in unstressed position (in English, unstressed vowels become **schwa** [ə]). Evidence of the influence of vowel reduction in native English learners of various foreign languages has been shown at the level of both perception and production.

Aldrich (2014) reports that SFL (Spanish as a Foreign Language) learners have problems in distinguishing stressed and unstressed /a/. The author speaks of centralization but, when taking into account that English's vowel inventory does not include a low central vowel, we note that vowel reduction implies two changes: front/back to central and low to mid. As a result, we should speak of *rising* instead of centralization. English has an open front vowel /æ/ and an open back vowel /ɑ/. Therefore, when producing the Spanish [a], learners should aim at a midpoint between them: an open vowel or the low central [ä]. Spanish native speakers also produce a slightly higher [a] in unstressed position; however, it is never produced as a mid-vowel.

Bland (2016) tackles the problem of vowel reduction by including duration as well. He studies the production of /e/ and /o/ by L1 English learners of Spanish and includes speech style and syllabic stress as independent variables in his study. His results show that /e/ and /o/ become more centralized and raised in unstressed positions, while also becoming shorter. He also points out that these vowels are more peripheral in formal speech style (this is a general effect that has also been noted for other languages like Chinese, see Yeh 2017). Cobb and Simonet (2015) also report centralization in unstressed positions. Moreover, they show that /u/ is produced as more fronted by L1 English speakers than by L1 Spanish speakers.

As previously mentioned, Spanish speakers do produce stressed and unstressed vowels differently; however, these differences are relatively minor. Figure 1.11, adapted from Martínez Celdrán (1984, 299), illustrates the centralization of

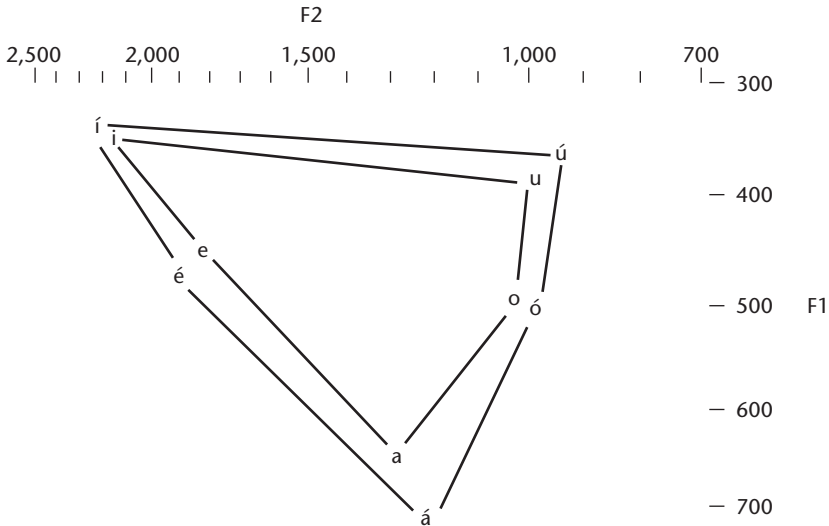


FIGURE 1.11 Formant chart depicting the centralization of unstressed vowels produced by L1 Spanish speakers.

unstressed vowels in Spanish, that is, in this figure, stressed vowels are more peripheral than unstressed vowels. Therefore, while it is the case that Spanish unstressed vowels are slightly centralized, stressed versus unstressed differences are not perceptible by the human ear.

Despite the great attention that has been drawn to vowel reduction, there are other salient differences between the English and Spanish vowel systems that cause difficulties for L1 English learners of Spanish. Fox, Flege, and Munro (1995) showed through multidimensional scaling that, whereas the Spanish vowel system could be described through two dimensions, three dimensions are needed in order to accurately classify the vowels of English. This is due to the different number of vowel phonemes in Spanish and English inventories, which can result in mismatches. Examples of mismatches can be observed in Morrison (2003), who studied how the perception of Spanish vowels by L1 English speakers affected their production. The author found that listeners were able to match Spanish /i/, /e/, and /o/ to the English /i/, /e/, and /o/ vowel categories, respectively, but perceptually mapped /a/ and /u/ to multiple L1 categories. This led the speakers to use English /i/, /e/, /æ/, /o/, and /u/ for Spanish /i/, /e/, /a/, /o/, and /u/, respectively. The author also noticed that the centers of vowel dispersion did not correspond to the center of any Spanish vowel.

A couple of final challenges to mention are that L1 English speakers tend to lengthen and diphthongize both final vowels and stressed vowels in Spanish (Díaz and Simonet 2015). They also have trouble discriminating some monophthongs and falling diphthongs (e.g., the difference between *pena* and *peina*) (Boomershine 2013; Romanelli and Menegotto 2015).

As for approaches to teaching Spanish vowels, it has been shown that successful results are achieved when learners participate in two-way immersion training (i.e., having native Spanish speakers in their immediate context in their own country, Menke 2010) or when they study abroad (Stevens 2011); however, according to Menke (2010) and Stevens (2011), students only benefit from such experiences at intermediate and high levels. At low proficiency levels, it appears that continuous exposure to Spanish has no effects (Menke 2010). The results of Reeder's (1999) study on short-term immersion trend in the same direction, he found that beginning and intermediate students did not show clear evidence of acquiring any of the Spanish vowels. On the other hand, advanced students achieved native-like performance in parts of Spanish's vowel system.

1.5 Teaching Spanish vowels

Everything that has been explained to this point has aimed to provide the reader with a foundational understanding of the pronunciation of Spanish vowels, which, in turn, can make it easier to teach or learn them. In this section, we offer a series of techniques for teaching Spanish vowels, all of which can be adapted based on the L1 of students.

It is widely known that students' mother tongue is crucial factor that influences L2 phonological acquisition. Non-native speakers whose native language is typologically different from the target language have more difficulties than those that come from a similar language. The main reason for this is that non-native speakers apply knowledge of the L1 when using the L2, resulting in language transfer. When the L1 and L2 share the same characteristics, the result is a great amount of positive transfer; however, this does not always apply to **phonology** since it is easier to create distinct perceptual categories for new sounds than for those that learners already have in their L1. This is why most teachers prefer preparing materials for students who share the same L1, and therefore, the same set of difficulties. On the other hand, SFL practitioners sometimes need to confront a classroom full of learners from very different origins, which causes difficulties when having to choose which phonemes to focus on (especially at lower levels). Overall, due to variation in the types of learners in Spanish language classrooms, we propose some general guidelines that are flexible enough to be adapted to each teacher's classroom reality.

1.5.1 Teaching vowel pronunciation in SFL

A key component of phonological competence is the development of skills related to the perception and production of vowel phonemes and their realizations in particular contexts (i.e., allophonic variants). Although research papers (Elliott 1997; Llisterri 2003; Lord 2005) and books (Poch Olivé 1999; Gil Fernández 2005) focusing on teaching Spanish pronunciation do exist, and pronunciation instruction is included on some course syllabi, commercial course books rarely include activities

to teach it. One reason for this is that Spanish pronunciation is considered a straightforward skill (Stevens 2006), since written and spoken forms generally reflect one another (as opposed to languages like English and French).

With regard to vowels, the reduced inventory of Spanish's system in contrast to that of other languages feeds the idea that they do not need to be taught. This has caused SFL teachers to not consider vowels as being problematic. As such, there is very little literature about on the topic. Moreover, the communicative approach has caused that, in many cases, teachers regard pronunciation training as repetitive and of little interest to students (Padilla-García 2007). A quick search about teaching materials related to vowel pronunciation in English and Spanish verifies that learners of English show greater concern about this topic than those studying SFL. Indeed, SFL students usually rate pronunciation as the least of their concerns in self-assessment and feedback tasks (Stevens 2006).

Learning an L2 with a smaller vowel inventory than that of the L1 may be easier, therefore, learners with smaller L1 vowel inventories than that of Spanish (e.g., Arabic or Quechua) will have more trouble than learners with larger L1 vowel inventories. On the other hand, even a language with a rich phonological system can have vowel differences that cause trouble for students; for example, in Brazilian Portuguese, which is typologically very close to Spanish, certain cases of open vowels can cause negative transfer in Brazilian learners of Spanish (Bazzan 2005).

Thinking of pronunciation as a straightforward skill is a changing tendency. In recent years, growing interest in the acquisition of Spanish vowels, often based on level of L1 interference, can be inferred from the increasing number of PhD dissertations and MA theses on the topic, which include work on L1 Portuguese and Chinese, among others (Cobb 2009; Santos 2014; Yeh 2014).

Despite common beliefs, inaccurate pronunciation is detrimental to comprehensibility and spoken and perceived fluency. The following subsections deal with pronunciation goals as a function of learners' proficiency level and common vowel mispronunciations.

1.5.2 Teaching vowels according to the learner's level

Language assessment standards differ depending on the language and the place where one is studying. Some well-known language certificates for English are TOEFL (USA) and Cambridge ESOL for English (UK), while CIEP is the expeditor institution for DELF and DALF (French), Goethe-Institut serves this role for German, and HSK Center of Beijing Language and Culture University does so for Chinese. The main language assessment body for European and American Spanish is the *Instituto Cervantes*.

Since the *Instituto Cervantes* evaluates students' knowledge following the benchmarks established by the Common European Framework of Reference for Languages, which was developed by the Council of Europe (CEFR), the following subsections develop vowel teaching proposals based on the learning level indicators proposed in Table 1.2 (Council of Europe 2011).

TABLE 1.2 Learning level indicators for pronunciation, extracted from CEFR (Council of Europe 2011)

<i>Phonological control</i>	
C	Can vary intonation and place sentence stress correctly in order to express finer shades of meaning.
B2	Has acquired a clear, natural pronunciation and intonation.
B1	Pronunciation is clearly intelligible even if a foreign accent is sometimes evident and occasional mispronunciations occur.
A2	Pronunciation is generally clear enough to be understood despite a noticeable foreign accent, but conversational partners will need to ask for repetition from time to time.
A1	Pronunciation of a very limited repertoire of learnt words and phrases can be understood with some effort by native speakers used to dealing with speakers of his/her language group.

1.5.3 Common challenges for learners

Phonetic inaccuracies due to negative transfer can be broadly classified according to error gravity. While aspects of pronunciation (and other features of spoken language) that are most detrimental to comprehensibility have been discussed in depth (Saito, Trofimovich, and Isaacs 2016), we apply a classification according to the seriousness of errors. The first type consists of mispronunciations that compromise the intelligibility of an utterance. They are acceptable at beginner levels (A1 and A2). The second type is mispronunciations that do not compromise intelligibility but are perceived as a clear indicator of “foreign accent” by native speakers. This type of mispronunciation is acceptable at intermediate levels (B1 and B2). The most common challenges related to vowels can be assigned to one of these two groups, as seen in Table 1.3.

1.5.4 Exercises for teaching vowels

In this subsection, some pronunciation exercises adapted to focus on the production of vowels are presented.

In L2 teaching, new content (e.g., grammatical forms) is introduced gradually when it is necessary to perform a new task. Therefore, most teaching guidelines, including the CEFR, establish a flexible schedule of new content that depends directly on the subject unit; for example, Spanish subjunctive forms are only taught when their use is required by the functional framework at hand (e.g., learning to express opinion and probability, *No creo que llueva*). However, pronunciation skills are required from the moment learners begin to speak and this makes pronunciation an early learning objective that, paradoxically, is often assumed as a long-term objective. As reported above, certain mispronunciations are acceptable at beginner levels if they do not interfere with utterance intelligibility, this suggests that although mastering pronunciation can be understood as a long-term objective, its teaching must begin at the same time as that of other linguistic skills. In order to speed up

TABLE 1.3 Common challenges for SFL learners. V stands for any vowel. The list is not exhaustive

	<i>Examples</i>	<i>Explanation</i>	<i>L1 that can cause the transfer</i>	<i>Examples</i>
Type 1	/V/ > [ə] /V/ > [u]	Vowel reduction. Consists of producing reduced vowels in unstressed positions.	English, Dutch, Russian, Bulgarian, among others.	[pe. 'le.le]> [pə. 'le.lə] [po. 'lo.lo]> [pu. 'lo.lu]
	/V/ > none	Elision of word final vowels.	Portuguese	['ka.sa]> ['kas]
	/V/ > none	Elision of unstressed vowels.	Portuguese	['loʝr.ðes]> ['loʝrðs]
	[e]>[i] [i]>[e] [o]>[u][u]>[o]	Confusion within anterior and posterior vowels.	Arabic Arabic	['me.sa]> ['mi.sa] ['to.no]> ['tu.nu]
	[-o] > [-ou] [-e] > [-ei] [u] > [y]	Diphthongization of final mid vowels. Change in place of articulation to compensate for rounding.	English French	['pe.na]> ['peɪ.na] [a. 'θul]> [a. 'θyl]
	[V] > [Ṽ]	Nasalization of vowels that are not surrounded by nasals.	French, Portuguese	['pan]> ['pân]
Type 2	['V] > ['V:]	Lengthening in stressed syllables.	Italian	[mã. 'jã.na]> [mã. 'jã:.na]
	[i+V] > [i+V] [u+V] > [u+V]	Problems producing diphthongs correctly due to tone transfer.	Mandarin Chinese	['pi.e]> ['pi.e] ['fue]> ['fu.e]
	[e] > [ɛ] [o] > [ɔ]	Production of open vowels.	Brazilian Portuguese	['bɛr.ðe]> [bɛr.ðe] [a. 'mɔr]> [a. 'mɔr]

the learning process, we consider it advantageous to develop a teaching strategy that capitalizes on student knowledge (Oliver Vera 2003). In order to do so, we propose dividing pronunciation exercises into three stages: 1. Awakening of vowel awareness, 2. Perceptual tasks, and 3. Production tasks.

1.5.4.1 Awakening of vowel awareness

Our first suggested exercise is intended for learners in their beginning stages. The exercise consists of two phases: recording and assessment. Students will be asked to produce and record L2 utterances. Then, teachers will provide students with native speaker materials and the class will analyze phonetic similarities and differences between their own productions and those of native speakers. The exercise's objective is to help students create new learning scaffolding.

1.5.4.2 Perception

1.5.4.2.1 Minimal pairs

Although it is usually reduced to a short listening exercise included in a textbook, training in **minimal pairs**, or pairs of words that differ minimally by a single sound (e.g., *pero* versus *paro*), is a common practice in L2 teaching. These exercises are usually strictly perceptual; however, students' performance improves when working in pairs/groups (i.e., with interaction) given such interaction forces them to produce the sound instead of only listening to it. One issue teachers should be mindful of is that these interactions can also lead to copying mispronunciations made by classmates. By increasing speaking practice opportunities, teachers can promote and encourage target-like pronunciation, as well as guide students toward an awareness of non-target-like forms. As such, it is advisable to include language production tasks together with perceptual exercises (see Section 1.5.4.3.1 for more ways of using minimal pairs to work on production). One such example is working with flash cards, where Student A reads a word, Student B chooses the word from a set of cards, and then they exchange roles. Then, students can choose some words and record sentences in which they appear. To illustrate, we present phonological oppositions through the following minimal pairs: *rosa/risa*, *mesa/masa*, *muro/mero*, *pero/poro*, *piiso/peso*, and *puro/paro*.

1.5.4.2.2 Spontaneous speech

The principal aim of learning a language is being competent in “real world” conversation. In order to train students to perceive and produce L2 vowels in real-life communicative contexts, they should not only be exposed to activities such as minimal pairs recorded in lab conditions, but also to conditions with background noise. In other words, students must be trained gradually in understanding speech in noisy environments, which makes them direct their attention to meaning and not to phonological form. If teachers wish to incorporate some attention to form, they could choose contents with examples of minimal pairs or use tasks in which students focus on determining the origin of speakers they hear. In order to gradually increase a focus on meaning, we propose using different media resources arranged by difficulty; for example, a radio interview will be easier than a television interview recorded in the streets, television programs or film scenes without background music will be easier than those with it, and spoken language is easier than listening to songs. Nevertheless, students that take official DELE or SIELE exams need to be exposed to all types of media. When selecting materials, teachers must take into account that, ideally, several dialectal varieties of Spanish should be included. This is especially true when preparing students for DELE exams. Using these materials to practice perception can be done through identification tasks (e.g., selecting words containing /i/).

1.5.4.3 Production

Ideally, in production tasks, self-assessment should be given preference over explicit correction. In order to do so, students can record L2 utterances and then (self-) evaluate them in conjunction with their teacher. Along these lines, it has been shown that L2 learners can benefit from hearing and, interestingly, visualizing their mistakes by means of acoustic analysis software such as *Praat* (Gorjian, Hayati, and Pourkhoni 2013, for an example of another software that can be used for self-evaluation and correction, see Morgan and García 2016); however, this only seems to be feasible at the university level, where students may take pronunciation or **phonetics** classes (and sometimes even receive training in acoustic analysis). Course syllabi in language schools do not include enough time to train students in reading **spectrograms** and the majority of students lack the required background knowledge to perform speech analysis.

1.5.4.3.1 Repetition and imitation

Foreign accent imitation can serve to ease the challenges associated with the pronunciation of new sounds (Markham 1997). Regarding perception tasks, it is useful to ask students to compare the vowels in their own native language to those of Spanish. Perceptual assimilation tasks or rated dissimilarity tasks can also be useful. In order to implement these types of practice, students can record sentences in their native language and compare their productions to similar recordings of native speakers of the L2. After this, students can record themselves imitating a Spanish accent in their own L1. Finally, they can try to imitate a native Spanish speaker producing Spanish words.

One of the most frequently used exercises to practice pronunciation is the repetition of minimal pairs. A list of possible minimal pairs, which expands upon the list in Section 1.5.4.2.1, is: *casa/cosa*, *casa/caso*, *quiso/queso*, *cara/cura*, *pera/pira*, *perra/porra*, *burra/borra*, *burro/burra*, *poco/Paco*, and *teja/teje*. In order to make students think about the consequences of mispronunciations of vowels, we propose an exercise where they need to look for a minimal pair partner for the following words: *cama*, *mesa*, *piso*, *poco*, *pura*, *caso*, *ame*, *pena*, *cambio*, and *viva*. Once they make their list of minimal pairs, they can go over the changes in meaning created by substituting one vowel for another in each pair.

Stress also produces minimal pairs in Spanish. In order to practice stress-based modifications to meaning, students can focus on shifting syllabic prominence in pairs such as: *lavo/lavó*, *canto/cantó*, *peso/pesó*, *ceno/cenó*, *grito/gritó*, *tiro/tiró*, *gozo/gozó*, *toco/tocó*, *lucho/luchó*, and *cruzo/cruzó*.

One of the most salient problems that we have addressed is vowel reduction. In order to suppress and overcome phonological processes that occur in learners' L1s, we propose two exercises. First, have students produce the following words while trying to ensure that all vowels within each word sound similar: *patata*, *Petete*, *requetén*, *raspa*, *caspa*, *parada*, *metete*, *pelele*. Second, a teaching strategy that can be

adopted when working with vowel reduction is to disguise sentences in order to make all syllables look like isolated words, and to make all mid vowels look stressed by writing an orthographic accent on them. Students read the resulting sentences, and when they achieve a sentence without reduction, they can read the well-written sentence while trying to maintain the quality of all vowels; for example, a sample preliminary input would be *Vi vó én un pué ble ci tó*, which would be followed the second input, *Vivo en un pueblecito*. Similarly, the activity could start with *La vi da és bé lla* and move on to *La vida es bella*.

Finally, vowel reduction is intrinsically related to rhythm, which can be tracked with a metronome. In class, students can repeat sentences together by following a metronome's movements. Since Spanish is a **syllable-timed** language, each time the metronome goes one way, students should say one syllable. Input sentences can be written with dashes in order to facilitate the task (e.g., *La-vi-da-es-be-lla*).

1.5.4.3.2 Games

There are several games that can serve as methods of practicing vowels. *Hangman* is a game frequently used in foreign language teaching and can be adapted to serve as a formal activity (i.e., where production is very controlled). In order to do so, the game can be played only with vowels. In this way, students are forced to repeat isolated vowels when guessing words and, subsequently, produce them in context when guessing words; for example, if a class starts with *t_ _ mp_*, each student needs to articulate vowels that his/her classmates can understand and later produce in order to complete word, which, in this case, is *tiempo* ['tjem.po]).

In the *Game of Growing Sentences*, Student 1 starts with one word (e.g., *casa*), Student 2 follows by adding another word (e.g., *casa grande*), Student 3 continues the phrase (e.g., *casa grande alta*), and so on and so forth until a meaningful sentence is completed. Teachers can intervene to properly link words in order to make sentences grammatically correct. Producing entire sentences that are formed through this game is another useful drill.

In the *Naming Game*, teachers ask students to think of words that contain 1, 2, 3, 4, or 5 vowels and that describe objects in the classroom. In groups, students locate objects that fulfill each of the five vowel number requirements. Then, each group shares its findings with the rest of the class and they are assessed, for instance, *luz* (1), *libro* (2), *pizarra* (3), *cartulina* (4), and *(mapa) geográfico* (5).

1.6 Conclusion

A general assumption is that Spanish's pronunciation system is "easy" compared to those of other languages. This mainly results from the fact that most **graphemes** correspond to only one phoneme. The production of vowels is included in this assumption; however, in this chapter, we showed that Spanish vowels are phonetically different from those of other languages, including those that are typologically close. Moreover, Spanish lacks many phonetic implementation rules that affect a

great number of languages; for example, vowel reduction, which is common in other languages and often causes negative transfer in learners. In order to equip teachers with some tools that can be used to help learners reduce the influence of factors such as L1 transfer, we proposed a series of perception and production exercises. Although mastering pronunciation (if it happens) is not expected until higher levels of proficiency, it needs to be understandable even at lower levels. As such, we urge teachers to consider including (vowel) pronunciation in a wider range of course syllabi and hope to have provided them with a point of departure to do so.

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